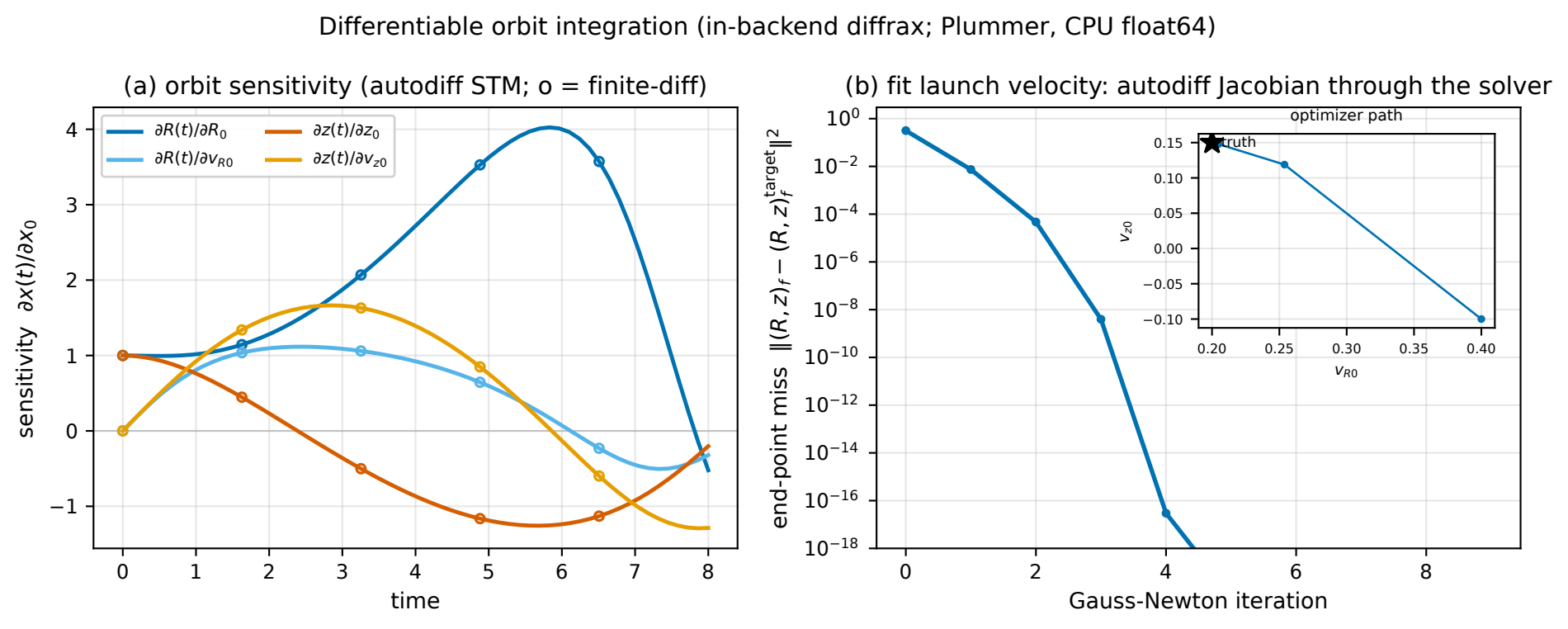
Differentiable orbit integration (in-backend diffrax; Plummer, CPU float64)
(a) orbit sensitivity (autodiff STM; o = finite-diff)
$\partial R(t)/\partial R_0$
$\partial R(t)/\partial v_{R0}$
$\partial z(t)/\partial z_0$
$\partial z(t)/\partial v_{z0}$
sensitivity $\partial x(t)/\partial x_0$
time
(b) fit launch velocity: autodiff Jacobian through the solver
optimizer path
truth
v_{z0}
v_{R0}
end-point miss $\|(R,z)_f - (R,z)_f^{\mathrm{target}}\|^2$
Gauss-Newton iteration